60. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
P, Q, R and S appeared in a test.
Question: Has P scored more marks than Q?
Statement–I: The sum of the marks scored by P and Q is equal to the sum of the marks scored by R and S.
Statement–II: The sum of the marks scored by P and S is more than the sum of the marks scored by Q and R.
Which one of the following is correct in respect of the above Question and the Statements?
(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.

Ans. (d) : Question If p is greater than Q
statement 1: The sum of marks scored by p and q is equal to marks scored by R and s.
$p + q = R + S$ ..... eq (1)
So statement 1 is not sufficient to answer the question-
Statement 2: The sum of marks scored by P&S more than sum of marks scored by Q and R $P + S > Q + R$ ..... eq (2)
So, statement 2 is also not sufficient to answer the question.
combining both statement,
Adding both equation,
$P + Q + P + S > R + S + Q + R$
$2P > 2R$
$P > R$
But this does not determine if p is greater than Q or not. Hence the question cannot be answered using both statement.

Directions for the followng 3 (three) items:
Read the following three passages and answer the items that follow the passage. Your answers to these items should be based on the passage only.

Passage–1

When an international team of scientists pumped a carbon dioxide and water mix into underground basalt rocks, basic chemistry took over. The acidic mixture dissolved rocks' calcium and magnesium and formed limestone. Basically carbon dioxide is converted into stone, exclaimed the scientists.

61. Which one of the following statements best reflects the most logical, rational and practical suggestion implied by the passage?
(a) It is a cheap and practical method to produce limestone at commercial level for building purposes.
(b) This can be used as one of the methods of carbon sequestration.
(c) Basalt rock can be made a good source of calcium and magnesium minerals by this methods.
(d) Good rock-dissolving acid can be produced by mixing carbon dioxide and water..

Ans. (b) : Options (a) is not correct as the passage talks about the chemical conversion of carbon dioxide into rock while options (a) discusses the method and cost of commercial production of limestone.
- Options (b) is the correct answer. The passage talks about carbon dioxide being converted into limestone by a chemical process. Options (b) supports this. It talks about a method of converting atmospheric carbon dioxide into limestone by capturing and storing it, that is, carbon sequestration, which was also proved by scientists.
- Options (c) is not correct as it talks about basaltic rocks being a good source of calcium and magnesium minerals while the passage says that a mixture of carbon dioxide and water dissolved calcium and magnesium present in basaltic rocks to form limestone. It nowhere talks about the extraction of calcium and magnesium from basaltic rocks.
- Options (d) is not correct. It talks about the production of good rock-soluble acid form a mixture of carbon dioxide and water but the passage does not support this.

Passage–2

Geographers analyzed 175 satellite images of ocean colour, which is an indicator of phytoplankton productivity at the ocean's surface, and found that giant icebergs are responsible for storing up to 20 percent of carbon in the Southern Ocean. The researchers discovered that melting water from giant icebergs which contains iron and other nutrients, supports hitherto unexpectedly high levels of phytoplankton growth.

62. Based on the above passage, the following assumptions have ben made:
1. Giant icebergs have a bearing on primary productivity and food chains of the Southern Ocean.
2. Melting of giant icebergs can produce climate change effects and impact world fisheries.
Which of the assumptions given above is/are valid?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : Assumption (1) is valid. According to the passage, meltwater from giant icebergs contains rich nutrients that support the growth of phytoplankton. Phytoplankton are an important part of the primary productivity of the oceans. Marine fishes use it as food. Thus phytoplankton are the base of the marine food chain. Hence the lines mentioned in the passage discuss the primary productivity of the southern ocean and its impact on food chains.
Note– This assumption is likely invalid, however, because the excerpt focuses primarily on the effects of icebergs on phytoplankton productivity and carbon storage, without discussing broader food chain effects.
- Assumption (2) is incorrect because the passage states that giant icebergs help in the growth of phytoplankton, it does not explicitly discuss broader climate change impacts or the impact on world fisheries.

Passage–3

Evolution has endowed caterpillars that feed on corn with a unique ability, they can induce the plant to turn off its defence against insect predators. This helps caterpillars to eat more and grow faster. The agent that causes this effect is the caterpillar's faeces or frass. The find could throw new light on compounds associated with plant response to pathogens like fungi or bacteria.

63. Which one of the following statements best reflects the most logical, rational and practical message conveyed by the passage?
(a) Farmers can use caterpillars to feed on weeds in their crop fields/plantations.
(b) This finding can help in the development of clinically useful antimicrobial compounds.
(c) This finding can help in the development of organic, ecologically sustainable pesticides.
(d) Caterpillars can be genetically modified to be predators of the other plant pests.

Ans. (c) : The passage discusses how caterpillars influence plants to deactivate their defenses against predators, potentially leading to insights for developing organic and ecologically sustainable pesticides. Understanding the mechanism could help in developing pest control methods that are environmentally friendly and effective, aligning closely with the passage's focus on plants defenses and ecological implications.

64. $32^5 + 2^{27}$ is divisble by:
(a) 3 (b) 7 (c) 10 (d) 11

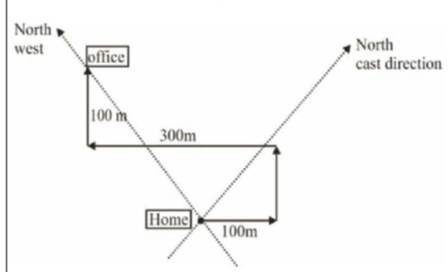
Ans. (c) : $32^5 + 2^{27}$
After simplify $\Rightarrow 2^{25} + 2^{27} = 2^{25}(1 + 2^2) = 2^{25} \times 5$
$= 2^{24} \times 2 \times 5 = 2^{24} \times 10$
Which is divisible by 10

65. Let p and q be positive integers satisfying $p < q$ and $p + q = k$. What is the smallest value of k that does not determine p and q uniquely?
(a) 3 (b) 4 (c) 5 (d) 6

Ans. (c) : Given–
P & q are positive integers,
$p < q$ and $p + q = k$
Smallest values of k, that does not determine p & q uniquely, – we have to check by options–
Let, $k = 3$ $p + q = k$ $p < q$
$1 + 2 = 3$ $1 < 2$
here p & q are uniquely
if $k = 4$, $p + q = k$ $p < q$
$1 + 3 =$ $1 < 3$
it's also give unique value.
if $k = 5$ $p + q = k$ $p < q$
$1 + 4 = 5$ $1 < 4$
or
$2 + 3 = 5$ $2 < 3$
here in this case, two situations may be and it is not uniquely.
There options c is correct.

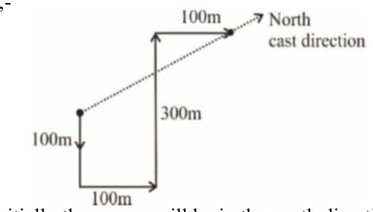
66. A person walks 100 m straight from his house, turns left and walks 100 m, again turns left and walks 300 m, then turns right and walks 100 m to reach his office. In which direction does he walk initially from his house if his office is exactly in the North-East direction?
(a) North-West (b) West
(c) South (d) South-West

Ans. (c) : According to given condition in question –



- But office is exactly in the north-East direction.
- We have to rotate the diagram by 90° in clock wise direction to satisfy the given condition in the question.

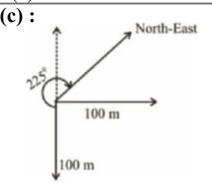
Now,-



So, initially the person will be in the south direction.

67. A person walks 100 m Westward, then turns left and walks 100 m. He then takes 225° turn clockwise. In which direction is he walking now?
(a) South-West (b) South-East
(c) North-West (d) North-East

Ans. (c) :



Form above figure we can say that the final direction would be North-East.

68. A Statement is given followed by two Conclusions numbered I and II. Consider the Statement and the Conclusions:
Statement: India is the world's largest producer of milk.
Conclusion–I: India is the world's largest exporter of milk.
Conclusion–II: India does not import milk.
Which one of the following is correct?
(a) Only Conclusion-I follows
(b) Only Conclusion-II follows
(c) Both Conclusion-I and Conclusion-II follow
(d) Neither Conclusion-I nor Conclusion-II follows

Ans. (d) : India is the world's largest producer of milk'. This statement is given, from this statement we can not say that India is the world's largest exporter of milk. It is not given in the statement. Hence this conclusion is not correct.
- India does not import milk. This conclusion is also incorrect because it is not mention in the statement.

69. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
Question: What are the values of m and n, where m and n are natural numbers?
Statement–I: $m + n > mn$ and $m > n$.
Statement–II: The product of m and n is 24.
Which one of the following is correct in respect of the above Question and the Statements?
(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.